JHU AI Engineering

Module 8 - PKI, SSL/TLS, Digital Certificates

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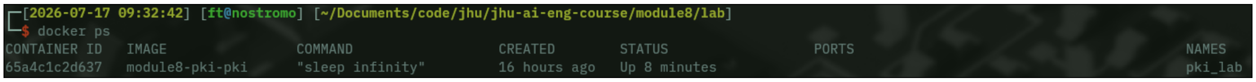
Part 1: Intro to OpenSSL toolkit

Note: I used `docker` to create the VM, not VMWare. Given this configuration, some differences in commands are necessary, but the outputs are more or less the same.



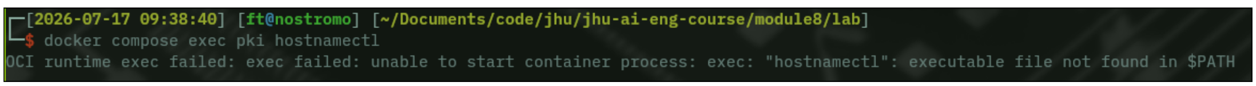
A screenshot of the running docker container.

Note: Given the setup using `docker`, we need to use `docker compose exec pki <command>` rather than the direct commands when operating directly inside the container.

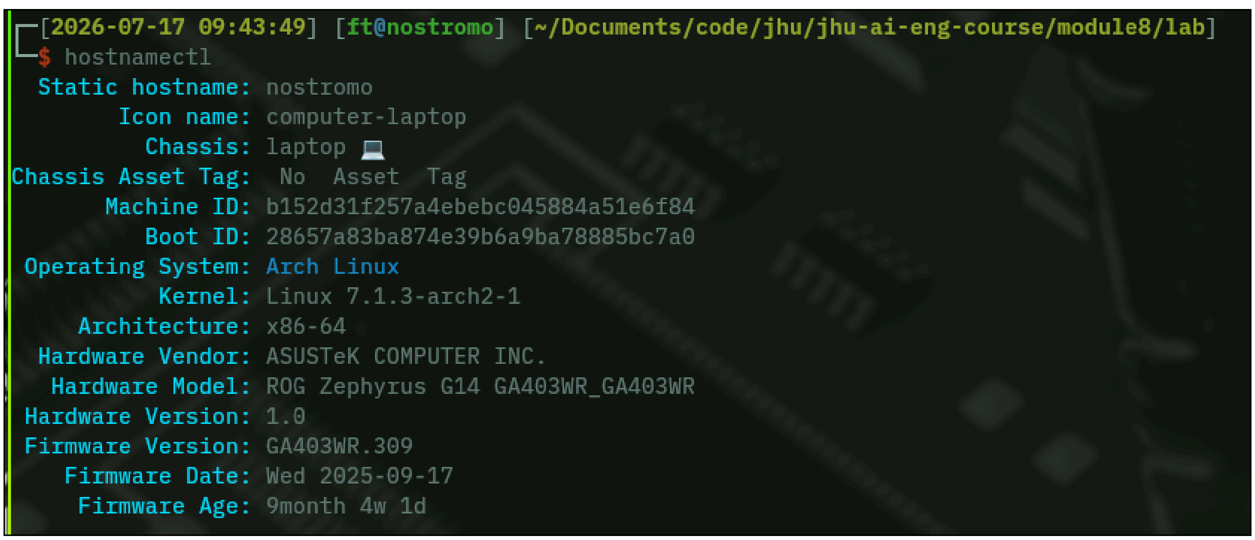
`hostnamectl`

This command fails given that:

a. A container is not a machine
b. Dockerized Ubuntu doesn't include `hostnamectl` in the minimal image.

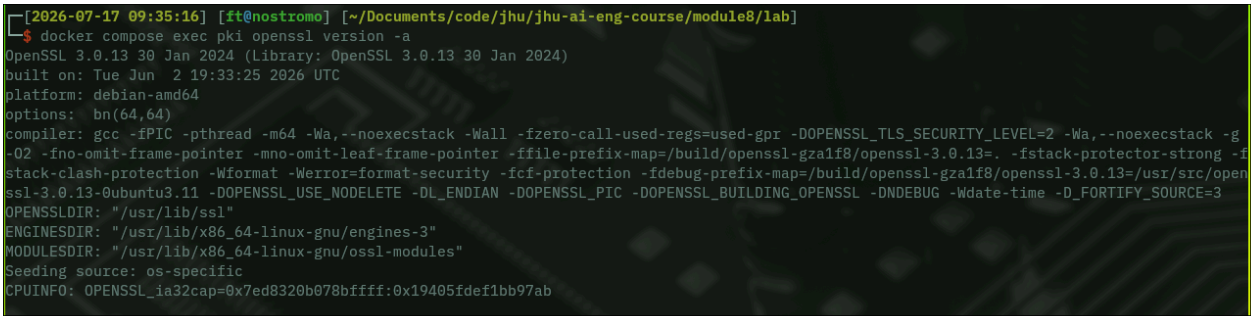


A screenshot of the failed `hostnamectl` running inside the container.



A screenshot of `hostnamectl` running on the native machine.

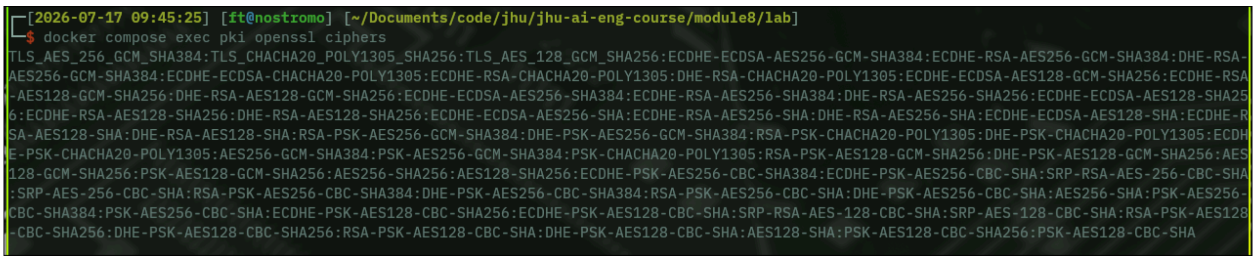
`openssl version -a`



```
┌[2026-07-17 09:35:16] [ft@nostromo] [~/Documents/code/jhu/jhu-ai-eng-course/module8/lab]
└$ docker compose exec pki openssl version -a
OpenSSL 3.0.13 30 Jan 2024 (Library: OpenSSL 3.0.13 30 Jan 2024)
built on: Tue Jun  2 19:33:25 2026 UTC
platform: debian-amd64
options:  bn(64,64)
compiler: gcc -fPIC -pthread -m64 -Wa,--noexecstack -Wall -fzero-call-used-regs=used-gpr -DOPENSSL_TLS_SECURITY_LEVEL=2 -Wa,--noexecstack -g
-O2 -fno-omit-frame-pointer -mno-omit-leaf-frame-pointer -ffile-prefix-map=/build/openssl-gza1f8/openssl-3.0.13=. -fstack-protector-strong -f
stack-clash-protection -Wformat -Werror=format-security -fcf-protection -fdebug-prefix-map=/build/openssl-gza1f8/openssl-3.0.13=/usr/src/open
ssl-3.0.13-0ubuntu3.11 -DOPENSSL_USE_NODELETE -DL_ENDIAN -DOPENSSL_PIC -DOPENSSL_BUILDING_OPENSSL -DNDEBUG -Wdate-time -D_FORTIFY_SOURCE=3
OPENSSLDIR: "/usr/lib/ssl"
ENGINESDIR: "/usr/lib/x86_64-linux-gnu/engines-3"
MODULESDIR: "/usr/lib/x86_64-linux-gnu/ossl-modules"
Seeding source: os-specific
CPUINFO: OPENSSL_ia32cap=0x7ed8320b078bffff:0x19405fdef1bb97ab
```

A screenshot of the `openssl version -a` run inside the container.

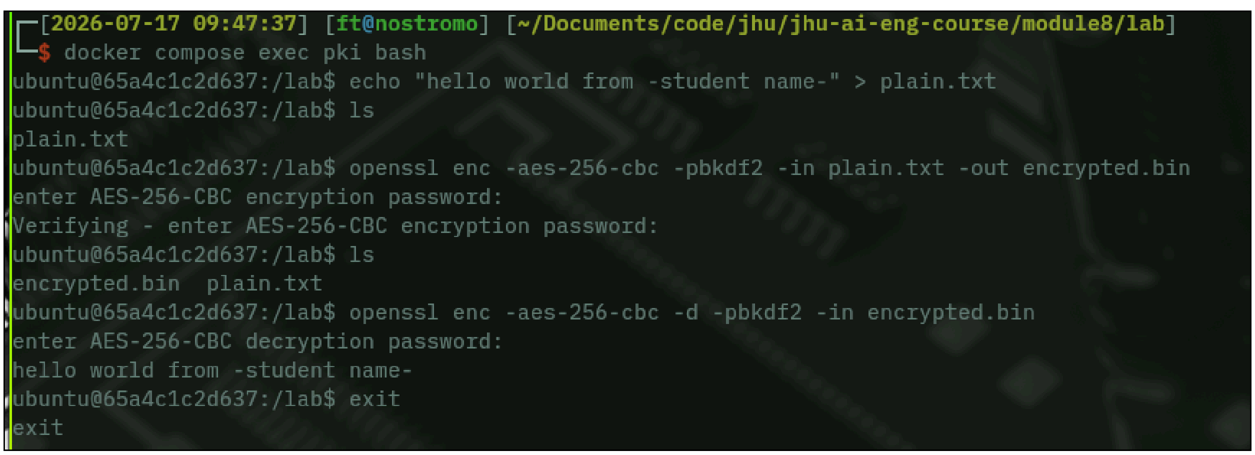
`openssl ciphers`



```
┌[2026-07-17 09:45:25] [ft@nostromo] [~/Documents/code/jhu/jhu-ai-eng-course/module8/lab]
└$ docker compose exec pki openssl ciphers
TLS_AES_256_GCM_SHA384:TLS_CHACHA20_POLY1305_SHA256:TLS_AES_128_GCM_SHA256:ECDHE-ECDSA-AES256-GCM-SHA384:ECDHE-RSA-AES256-GCM-SHA384:DHE-RSA-
AES256-GCM-SHA384:ECDHE-ECDSA-CHACHA20-POLY1305:ECDHE-RSA-CHACHA20-POLY1305:DHE-RSA-CHACHA20-POLY1305:ECDHE-ECDSA-AES128-GCM-SHA256:ECDHE-RSA
-AES128-GCM-SHA256:DHE-RSA-AES128-GCM-SHA256:ECDHE-ECDSA-AES256-SHA384:ECDHE-RSA-AES256-SHA384:DHE-RSA-AES256-SHA256:ECDHE-ECDSA-AES128-SHA25
6:ECDHE-RSA-AES128-SHA256:DHE-RSA-AES128-SHA256:ECDHE-ECDSA-AES256-SHA:ECDHE-RSA-AES256-SHA:DHE-RSA-AES256-SHA:ECDHE-ECDSA-AES128-SHA:ECDHE-R
SA-AES128-SHA:DHE-RSA-AES128-SHA:RSA-PSK-AES256-GCM-SHA384:DHE-PSK-AES256-GCM-SHA384:RSA-PSK-CHACHA20-POLY1305:DHE-PSK-CHACHA20-POLY1305:ECDH
E-PSK-CHACHA20-POLY1305:AES256-GCM-SHA384:PSK-AES256-GCM-SHA384:PSK-CHACHA20-POLY1305:RSA-PSK-AES128-GCM-SHA256:DHE-PSK-AES128-GCM-SHA256:AES
128-GCM-SHA256:PSK-AES128-GCM-SHA256:AES256-SHA256:AES128-SHA256:ECDHE-PSK-AES256-CBC-SHA384:ECDHE-PSK-AES256-CBC-SHA:SRP-RSA-AES-256-CBC-SHA
:SRP-AES-256-CBC-SHA:RSA-PSK-AES256-CBC-SHA384:DHE-PSK-AES256-CBC-SHA384:RSA-PSK-AES256-CBC-SHA:DHE-PSK-AES256-CBC-SHA:AES256-SHA:PSK-AES256-
CBC-SHA384:PSK-AES256-CBC-SHA:ECDHE-PSK-AES128-CBC-SHA256:ECDHE-PSK-AES128-CBC-SHA:SRP-RSA-AES-128-CBC-SHA:SRP-AES-128-CBC-SHA:RSA-PSK-AES128
-CBC-SHA256:DHE-PSK-AES128-CBC-SHA256:RSA-PSK-AES128-CBC-SHA:DHE-PSK-AES128-CBC-SHA:AES128-SHA:PSK-AES128-CBC-SHA256:PSK-AES128-CBC-SHA
```

A screenshot of the `openssl ciphers` run inside the container.

Part 2: Quick warm up on Symmetric Encryption and Decryption



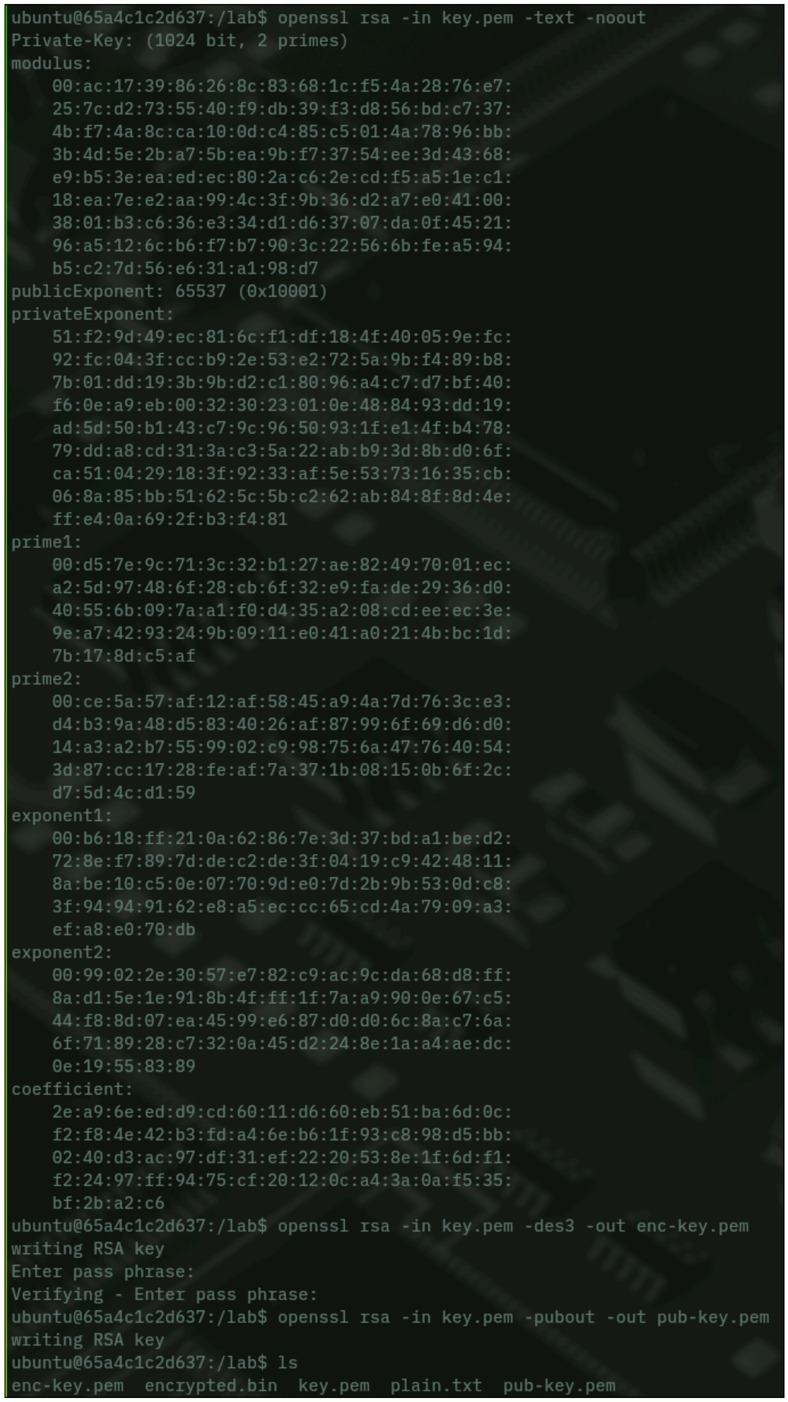
```
┌[2026-07-17 09:47:37] [ft@nostromo] [~/Documents/code/jhu/jhu-ai-eng-course/module8/lab]
└$ docker compose exec pki bash
ubuntu@65a4c1c2d637:/lab$ echo "hello world from -student name-" > plain.txt
ubuntu@65a4c1c2d637:/lab$ ls
plain.txt
ubuntu@65a4c1c2d637:/lab$ openssl enc -aes-256-cbc -pbkdf2 -in plain.txt -out encrypted.bin
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
ubuntu@65a4c1c2d637:/lab$ ls
encrypted.bin  plain.txt
ubuntu@65a4c1c2d637:/lab$ openssl enc -aes-256-cbc -d -pbkdf2 -in encrypted.bin
enter AES-256-CBC decryption password:
hello world from -student name-
ubuntu@65a4c1c2d637:/lab$ exit
exit
```

A screenshot of all steps required in Part 2. Demonstrating encrypting and decrypting a text file.

Part 3: PKI, Working with Public and Private Keys

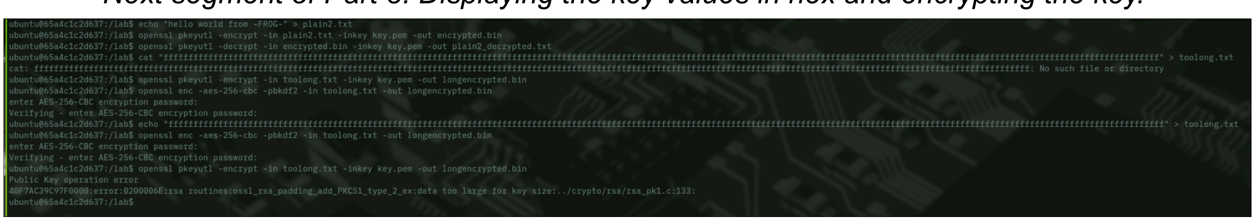
```
┌─[2026-07-17 11:33:45] [ft@nostromo] [~/Documents/code/jhu/jhu-ai-eng-course/module8/lab]
└─$ docker compose exec pki bash
ubuntu@65a4c1c2d637:/lab$ ls
encrypted.bin  plain.txt
ubuntu@65a4c1c2d637:/lab$ openssl genrsa -out key.pem 1024
ubuntu@65a4c1c2d637:/lab$ ls
encrypted.bin  key.pem  plain.txt
ubuntu@65a4c1c2d637:/lab$ cat key.pem
-----BEGIN PRIVATE KEY-----
MIICdwIBADANBgkqhkiG9w0BAQEFAASCAmEwggJdAgEAAoGBAKwXOYYmjINoHPVK
KHbnJXzSc1VA+ds589hWvcc3S/dKjMoQDcSFxQFKeJa7O01eK6db6pv3N1TuPUNo
6bU+6u3sgCrGLs31pR7BGOp+4qqZTD+bNtKn4EEAOAGzxjbjNNHWNwfaD0UhlqUS
bLb3t5A8IlZr/qWUtcJ9VuYxoZjXAgMBAAECgYBR8p1J7IFs8d8YT0AFnvyS/AQ/
zLkuU+JyWpv0ibh7Ad0ZO5vSwYCWpMfXv0D2DqnrADIwIwEOSIST3RmtXVCxQ8ec
llCTH+FPtHh53ajNMTrDWiKruT2L0G/KUQQpGD+SM69eU3MWNcsGioW7UWJcW8Ji
q4SPjU7/5AppL7P0gQJBANV+nHE8MrEnroJJcAHsol2XSG8oy28y6freKTbQQFVr
CXqh8NQ1ogjN7uw+nqdCkySbCRHgQaAhS7wdexeNxa8CQQDOWlevEq9YRalKfXY8
49SzmkjVg0Amr4eZb2nW0BSjordVmQLJmHVqR3ZAVD2HzBco/q96NxsIFQtvLNdd
TNFZAkEAthj/IQpihn49N72hvtJyjveJfd7C3j8EGclCSBGKvhDFDgdwneB9K5tT
Dcg/lJSRYuil7MxlzUp5CaPvqOBw2wJBAJkCLjBX54LJrJzaaNj/itFeHpGLT/8f
eqmQDmfFRPiNB+pFmeaH0NBsisdqb3GJKMcyCkXSJI4apK7cDhlVg4kCQC6pbu3Z
zWAR1mDrUbptDPL4TkKz/aRuth+TyJjVuwJA06yX3zHvIiBTjh9t8fIkl/+Udc8g
EgykOgr1Nb8rosY=
-----END PRIVATE KEY-----
```

Segment 1 of Part 3. Generating a key and displaying the value. Insecure!



```
ubuntu@65a4c1c2d637:/lab$ openssl rsa -in key.pem -text -noout
Private-Key: (1024 bit, 2 primes)
modulus:
    00:ac:17:39:86:26:8c:83:68:1c:f5:4a:28:76:e7:
    25:7c:d2:73:55:40:f9:db:39:f3:d8:56:bd:c7:37:
    4b:f7:4a:8c:ca:10:0d:c4:85:c5:01:4a:78:96:bb:
    3b:4d:5e:2b:a7:5b:ea:9b:f7:37:54:ee:3d:43:68:
    e9:b5:3e:ea:ed:ec:80:2a:c6:2e:cd:f5:a5:1e:c1:
    18:ea:7e:e2:aa:99:4c:3f:9b:36:d2:a7:e0:41:00:
    38:01:b3:c6:36:e3:34:d1:d6:37:07:da:0f:45:21:
    96:a5:12:6c:b6:f7:b7:90:3c:22:56:6b:fe:a5:94:
    b5:c2:7d:56:e6:31:a1:98:d7
publicExponent: 65537 (0x10001)
privateExponent:
    51:f2:9d:49:ec:81:6c:f1:df:18:4f:40:05:9e:fc:
    92:fc:04:3f:cc:b9:2e:53:e2:72:5a:9b:f4:89:b8:
    7b:01:dd:19:3b:9b:d2:c1:80:96:a4:c7:d7:bf:40:
    f6:0e:a9:eb:00:32:30:23:01:0e:48:84:93:dd:19:
    ad:5d:50:b1:43:c7:9c:96:50:93:1f:e1:4f:b4:78:
    79:dd:a8:cd:31:3a:c3:5a:22:ab:b9:3d:8b:d0:6f:
    ca:51:04:29:18:3f:92:33:af:5e:53:73:16:35:cb:
    06:8a:85:bb:51:62:5c:5b:c2:62:ab:84:8f:8d:4e:
    ff:e4:0a:69:2f:b3:f4:81
prime1:
    00:d5:7e:9c:71:3c:32:b1:27:ae:82:49:70:01:ec:
    a2:5d:97:48:6f:28:cb:6f:32:e9:fa:de:29:36:d0:
    40:55:6b:09:7a:a1:f0:d4:35:a2:08:cd:ee:ec:3e:
    9e:a7:42:93:24:9b:09:11:e0:41:a0:21:4b:bc:1d:
    7b:17:8d:c5:af
prime2:
    00:ce:5a:57:af:12:af:58:45:a9:4a:7d:76:3c:e3:
    d4:b3:9a:48:d5:83:40:26:af:87:99:6f:69:d6:d0:
    14:a3:a2:b7:55:99:02:c9:98:75:6a:47:76:40:54:
    3d:87:cc:17:28:fe:af:7a:37:1b:08:15:0b:6f:2c:
    d7:5d:4c:d1:59
exponent1:
    00:b6:18:ff:21:0a:62:86:7e:3d:37:bd:a1:be:d2:
    72:8e:f7:89:7d:de:c2:de:3f:04:19:c9:42:48:11:
    8a:be:10:c5:0e:07:70:9d:e0:7d:2b:9b:53:0d:c8:
    3f:94:94:91:62:e8:a5:ec:cc:65:cd:4a:79:09:a3:
    ef:a8:e0:70:db
exponent2:
    00:99:02:2e:30:57:e7:82:c9:ac:9c:da:68:d8:ff:
    8a:d1:5e:1e:91:8b:4f:ff:1f:7a:a9:90:0e:67:c5:
    44:f8:8d:07:ea:45:99:e6:87:d0:d0:6c:8a:c7:6a:
    6f:71:89:28:c7:32:0a:45:d2:24:8e:1a:a4:ae:dc:
    0e:19:55:83:89
coefficient:
    2e:a9:6e:ed:d9:cd:60:11:d6:60:eb:51:ba:6d:0c:
    f2:f8:4e:42:b3:fd:a4:6e:b6:1f:93:c8:98:d5:bb:
    02:40:d3:ac:97:df:31:ef:22:20:53:8e:1f:6d:f1:
    f2:24:97:ff:94:75:cf:20:12:0c:a4:3a:0a:f5:35:
    bf:2b:a2:c6
ubuntu@65a4c1c2d637:/lab$ openssl rsa -in key.pem -des3 -out enc-key.pem
writing RSA key
Enter pass phrase:
Verifying - Enter pass phrase:
ubuntu@65a4c1c2d637:/lab$ openssl rsa -in key.pem -pubout -out pub-key.pem
writing RSA key
ubuntu@65a4c1c2d637:/lab$ ls
enc-key.pem  encrypted.bin  key.pem  plain.txt  pub-key.pem
```

Next segment of Part 3. Displaying the key values in hex and encrypting the key.



```
ubuntu@65a4c1c2d637:/lab$ echo "hello world from -FROG-" > plain2.txt
ubuntu@65a4c1c2d637:/lab$ openssl pkeyutl -encrypt -in plain2.txt -inkey key.pem -out encrypted.bin
ubuntu@65a4c1c2d637:/lab$ openssl pkeyutl -decrypt -in encrypted.bin -inkey key.pem -out plain2_decrypted.txt
ubuntu@65a4c1c2d637:/lab$ cat "ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff" > toolong.txt
cat: ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff: No such file or directory
ubuntu@65a4c1c2d637:/lab$ openssl pkeyutl -encrypt -in toolong.txt -inkey key.pem -out longencrypted.bin
ubuntu@65a4c1c2d637:/lab$ openssl enc -aes-256-cbc -pbkdf2 -in toolong.txt -out longencrypted.bin
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
ubuntu@65a4c1c2d637:/lab$ echo "ffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffffff" > toolong.txt
ubuntu@65a4c1c2d637:/lab$ openssl enc -aes-256-cbc -pbkdf2 -in toolong.txt -out longencrypted.bin
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
ubuntu@65a4c1c2d637:/lab$ openssl pkeyutl -encrypt -in toolong.txt -inkey key.pem -out longencrypted.bin
Public Key operation error
40F7AC39C97F0000:error:0200006E:rsa routines:ossl_rsa_padding_add_PKCS1_type_2_ex:data too large for key size:../crypto/rsa/rsa_pk1.c:133:
ubuntu@65a4c1c2d637:/lab$
```

A demonstration of overflowing the RSA character limit, including a fumbled command!

Part 3.5: Simulating the Handoff to another classmate

```
┌[2026-07-17 11:55:47] [ft@nostromo] [~/Documents/code/jhu/jhu-ai-eng-course/module8/lab]
└$ docker compose exec pki bash
ubuntu@65a4c1c2d637:/lab$ ls
enc-key.pem  encrypted.bin  key.pem  longencrypted.bin  plain.txt  plain2.txt  plain2_decrypted.txt  pub-key.pem  toolong.txt
ubuntu@65a4c1c2d637:/lab$ openssl genrsa -out fellow.pem 1024
ubuntu@65a4c1c2d637:/lab$ ls
enc-key.pem  encrypted.bin  fellow.pem  key.pem  longencrypted.bin  plain.txt  plain2.txt  plain2_decrypted.txt  pub-key.pem  toolong.txt
ubuntu@65a4c1c2d637:/lab$ openssl rsa -in fellow.pem -pubout -out pub-fellow.pem
writing RSA key
ubuntu@65a4c1c2d637:/lab$ ls
enc-key.pem  encrypted.bin  fellow.pem  key.pem  longencrypted.bin  plain.txt  plain2.txt  plain2_decrypted.txt  pub-fellow.pem  pub-key.pem  toolong.txt
ubuntu@65a4c1c2d637:/lab$ vi fred.txt
bash: vi: command not found
ubuntu@65a4c1c2d637:/lab$ echo -e "Fred Teumer\nSteak and Waffles\nRally House Bops" > fred.txt
ubuntu@65a4c1c2d637:/lab$ cat fred.txt
Fred Teumer
Steak and Waffles
Rally House Bops
ubuntu@65a4c1c2d637:/lab$ openssl pkeyutl -encrypt -inkey fellow.pem -pubin -in fred.txt -out encryptedname.enc
Could not read public key from fellow.pem
404786A4297F0000:error:1608010C:STORE routines:ossl_store_handle_load_result:unsupported:../crypto/store/store_result.c:151:
pkeyutl: Error initializing context
ubuntu@65a4c1c2d637:/lab$ openssl rsa -in fellow.pem -des3 -out enc-fellow.pem
writing RSA key
Enter pass phrase:
Verifying - Enter pass phrase:
Verify failure
unable to write key
4057D7DD137F0000:error:1400006B:UI routines:UI_process:processing error:../crypto/ui/ui_lib.c:548:while reading strings
4057D7DD137F0000:error:07880028:common libcrypto routines:do_ui_passphrase:UI lib:../crypto/passphrase.c:187:
4057D7DD137F0000:error:1C80009F:Provider routines:p8info_to_encp8:unable to get passphrase:../providers/implementations/encode_decode/encode_key2any.c:116:
ubuntu@65a4c1c2d637:/lab$ openssl pkeyutl -encrypt -inkey pub-fellow.pem -pubin -in fred.txt -out encryptedname.enc
ubuntu@65a4c1c2d637:/lab$ openssl pkeyutl -decrypt -inkey fellow.pem -in encryptedname.enc -out decryptedname.txt
ubuntu@65a4c1c2d637:/lab$ cat decryptedname.txt
Fred Teumer
Steak and Waffles
Rally House Bops
ubuntu@65a4c1c2d637:/lab$
```

Simulating using a separate public key to encrypt the file, decrypting with a different private key.